

Supplementary Material: Reproducibility Details

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This document supplements the article “Boundary Information Geometry for S-box Side-Channel Leakage: Walsh Covariance, Active Fisher Structure, and CPA Belief Dynamics.” It records the computational protocol used for the numerical claims in the manuscript. The computations are intentionally small and exhaustive: for $s = 4$, all input values, all key hypotheses, all Walsh coordinates, and all active coordinate planes are enumerated. The accompanying repository archive is tagged `v1.0.5`.

1 Software Environment

The distributed scripts use Python 3 and standard numerical packages. The core dependencies are `numpy` for finite-array enumeration, Monte Carlo CPA simulation, and linear algebra, `scipy` for symmetric eigenspace computations, and `matplotlib` for figure generation. The deterministic tables are obtained by exhaustive enumeration. The CPA validation uses a fixed pseudorandom seed, 20260622, Gaussian noise standard deviation $\sigma = 2$, and 3000 trials per S-box. When an additional 4-bit bijection is used in `verify_curvature.py`, the script uses a fixed table explicitly listed in the source code.

2 Commands

The main numerical outputs can be regenerated from the project root with:

```
python gen_figures.py
python simulate_cpa.py
python compute_curvatures.py
python verify_curvature.py
```

The first command regenerates the Walsh-spectrum and local key-separation figures. The second command regenerates the CPA Monte Carlo figure and the CSV summary of success probabilities and key ranks. The third command constructs the boundary covariance matrix, projects onto the active eigenspace, computes the projected third cumulant contractions, and evaluates the curvature diagnostic. The fourth command is a sanity check for the value $K = -1/4$ on PRESENT and on an additional fixed random bijection.

3 Arithmetic and Precision

The Walsh coefficients, boundary covariance entries, and third cumulant entries are obtained from finite sums over $\{0, 1\}^4$. These quantities are therefore integer or rational before diagonalization. The eigenspace projection and the curvature table are evaluated in floating-point arithmetic after the exact finite sums have been constructed. The reported curvature values are reproducible numerical diagnostics for the tested S-boxes under the projection and normalization conventions used in the manuscript.

4 Sign Conventions

The Walsh transform convention is

$$\hat{S}(\alpha, \beta) = \sum_{a \in \{0,1\}^s} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a)}.$$

The active-eigenspace curvature diagnostic follows the convention used in Section 6 of the manuscript. Changing the Riemann tensor sign convention would change the sign of all reported sectional curvature values; the scripts and the manuscript use the same convention.

5 Reproducibility Coverage

The computational protocol covers the quantities reported in the paper:

- (i) Walsh spectra and absolute-coefficient histograms;
- (ii) boundary covariance matrices and active ranks/eigenvalues;
- (iii) local Walsh/Fisher key-separation proxies;
- (iv) Hamming-weight leakage descriptors and pairwise leakage gaps;
- (v) Monte Carlo CPA success probabilities and true-key ranks;
- (vi) projected active-eigenspace curvature diagnostics for PRESENT, GIFT, and SKINNY.

6 Files

File	Role
<code>gen_figures.py</code>	Regenerates the manuscript figures.
<code>simulate_cpa.py</code>	Regenerates CPA simulation figure and CSV results.
<code>cpa_simulation_results.csv</code>	Records simulated success probabilities and ranks.
<code>compute_curvatures.py</code>	Computes active ranks, eigenvalues, and curvature values.
<code>verify_curvature.py</code>	Performs an independent curvature sanity check.
<code>supplementary_README.md</code>	Lists commands and expected outputs.
<code>requirements.txt</code>	Lists the Python package dependencies.